COMP1013 Analytics Programming — Assignment Autumn 2026

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1 Data Loading and Preparation

```
# Load all required libraries
library(tidyverse)    # data wrangling (dplyr) and visualisation (ggplot2)
library(lubridate)    # date handling - year(), as.Date()
library(knitr)        # kable() for tables
library(kableExtra)   # enhanced table formatting (striped, hover styles)
```

```
# ------------------------------------------------------------------
# Load the three CSV datasets
# Ensure all three files are in the same working directory as this .Rmd
# ------------------------------------------------------------------
players  <- read.csv("players.csv",  stringsAsFactors = FALSE)
games    <- read.csv("games.csv",    stringsAsFactors = FALSE)
sessions <- read.csv("sessions.csv", stringsAsFactors = FALSE)

# Convert signup_date from string to R Date type for date arithmetic
players$signup_date <- as.Date(players$signup_date)
```

```
# Quick inspection of each dataset
cat("players  :", nrow(players),  "rows,", ncol(players),  "columns\n")
```

```
## players  : 5000 rows, 4 columns
```

```
cat("games    :", nrow(games),    "rows,", ncol(games),    "columns\n")
```

```
## games    : 50 rows, 4 columns
```

```
cat("sessions :", nrow(sessions), "rows,", ncol(sessions), "columns\n")
```

```
## sessions : 20000 rows, 5 columns
```

```
# Check for NA values in all three datasets
cat("\nNA counts - players:\n");  print(colSums(is.na(players)))
```

```
##
## NA counts - players:
```

```
##   player_id         age     country signup_date
##           0           0           0           0
```

```r
cat("\nNA counts - games:\n");    print(colSums(is.na(games)))
```

```
##
## NA counts - games:

##   game_id game_name     genre  platform
##         0         0         0         0
```

```r
cat("\nNA counts - sessions:\n"); print(colSums(is.na(sessions)))
```

```
##
## NA counts - sessions:

##        session_id         player_id           game_id play_time_minutes
##                 0                 0                 0                 0
##             score
##                 0
```

Luckily none of the three datasets featured any missing values so no imputation was required. I did keep both na.rm = TRUE and filter(!is.na(…)) throughout though just in case something surprising came up post-joining of the tables.

2 Task 1 — Player Engagement Analysis by Experience Level

2.1 Task Explanation

For this task I had to split players into three groups depending on when they signed up. Players who joined before 2023 are Veterans, those who joined during 2023 are Intermediate, and anyone who joined after 2023 is New. Once I had all the players I could find I calculated how many were in each group, what their average session length was in minutes and how their average score was. The notion was to see if being on the platform for longer made any difference in what ways people played.

2.2 Assumptions

As the only means of classifying experience level was the use of the signup year for each player - irrespective of their playing experience - players that signed up in 2021 but did not play would still be categorized as a Veteran. Similarly, any players whose signup year could not be determined were also left out of this data. Furthermore, each player's total number of game sessions experienced by that player is utilized in the calculation of each of these averages - irrespective of the year in which they occurred.

2.3 Data Wrangling

```r
# Step 1: Classify each player into an experience group based on signup year
players_exp <- players %>%
  mutate(
    signup_year      = year(signup_date),          # extract integer year
    experience_level = case_when(
      signup_year <  2023 ~ "Veteran",             # registered before 2023
      signup_year == 2023 ~ "Intermediate",        # registered in 2023
      signup_year >  2023 ~ "New",                 # registered after 2023
      TRUE                ~ NA_character_          # catch-all for unexpected NAs
    )
  ) %>%
  filter(!is.na(experience_level))                 # remove any unclassified rows

# Step 2: Join players with sessions on player_id (left join keeps all players)
player_sessions <- players_exp %>%
  left_join(sessions, by = "player_id")

# Step 3: Summarise key metrics per experience level
task1_summary <- player_sessions %>%
  group_by(experience_level) %>%
  summarise(
    num_players      = n_distinct(player_id),                         # unique player count
    avg_play_time_min = round(mean(play_time_minutes, na.rm = TRUE), 2), # avg session length
```

```r
    avg_score          = round(mean(score, na.rm = TRUE), 2)           # avg score per session
  ) %>%
  # Order factor levels logically: most to least experienced
  mutate(experience_level = factor(experience_level,
         levels = c("Veteran", "Intermediate", "New"))) %>%
  arrange(experience_level)
```

2.4 Results Table

```r
# Display results as a formatted kable table
task1_summary %>%
  kable(
    col.names = c("Experience Level", "Number of Players",
                  "Avg Play Time (min)", "Avg Score"),
    caption   = "Player Engagement Metrics by Experience Level",
    booktabs  = TRUE
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position"),
                full_width = FALSE)
```

Table 1: Player Engagement Metrics by Experience Level

Experience Level	Number of Players	Avg Play Time (min)	Avg Score
Veteran	1786	92.26	501.04
Intermediate	1851	91.73	498.20
New	1363	92.41	495.96

2.5 Visualisation

```r
ggplot(task1_summary,
       aes(x = experience_level, y = avg_play_time_min,
           fill = experience_level)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = avg_play_time_min),
            vjust = -0.4, size = 4.5, fontface = "bold") +
  scale_fill_manual(values = c("Veteran"      = "#2C7BB6",
                               "Intermediate" = "#ABD9E9",
                               "New"          = "#FDAE61")) +
  labs(
    title    = "Average Play Time by Player Experience Level",
    subtitle = "Grouped by year of account signup",
    x        = "Experience Level",
```

```
  y        = "Average Play Time (minutes)"
) +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold"))
```

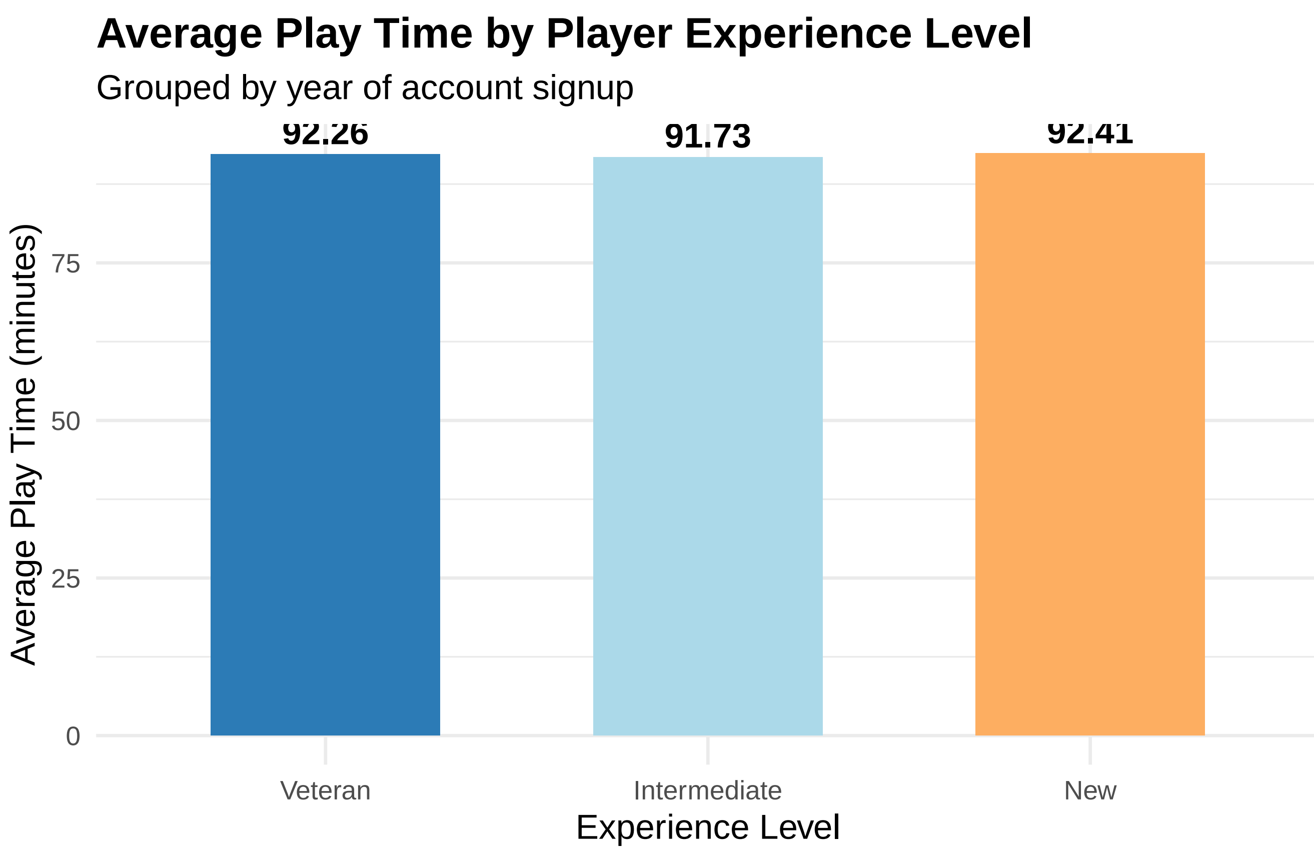

Figure 1: Average Play Time by Player Experience Level

2.6 Interpretation

The results were a bit surprising as all three groups featured nearly exactly the same amount of playing time per session (around 91-92 minutes) and very similar scoring outcomes (496-501). The veterans of the game do indeed edge out new players in average score - which makes sense given experience - but again the difference is so slight as to be effectively nothing. This does tell me, however, that new players are neither struggling to keep up nor dropping off participation - they are just as engaged as those with longer experience levels. This is another great finding for Play Pulse Studios as it shows that the platform is successful in maintaining the attention of the users, regardless of the length of time that they have been on the platform for.

3 Task 2 — Game Genre Analysis

3.1 Task Explanation

This task looked at the popularity of each game genre within the platform. There are five genres - Action, Puzzle, RPG, Sports and Strategy. For each of these I worked out the average play time, average score, total number of sessions and how many unique players played that genre. This provides a picture of which game genres people are drawn to and which don't receive such love.

3.2 Assumption

To get the genre for each session I joined the sessions table with the games table using the game_id. Any sessions that did not have a game associated with them were dropped since I did not know the genre of those games. The measure of popularity was primarily the unique player count but also looked at play time and number of sessions as additional measurements of popularity.

3.3 Data Wrangling

```r
# Step 1: Join sessions with games to get genre for every session
sessions_genres <- sessions %>%
  left_join(games, by = "game_id") %>%
  filter(!is.na(genre))           # drop any sessions without a matched genre

# Step 2: Summarise engagement metrics per genre
task2_summary <- sessions_genres %>%
  group_by(genre) %>%
  summarise(
    avg_play_time      = round(mean(play_time_minutes, na.rm = TRUE), 2),
    avg_score          = round(mean(score, na.rm = TRUE), 2),
    num_sessions       = n(),                       # total session count
    num_unique_players = n_distinct(player_id)      # distinct players per genre
  ) %>%
  arrange(desc(num_unique_players))                 # sort by popularity
```

3.4 Results Table

```r
task2_summary %>%
  kable(
    col.names = c("Genre", "Avg Play Time (min)", "Avg Score",
                  "Sessions", "Unique Players"),
    caption   = "Player Engagement Metrics by Game Genre",
    booktabs  = TRUE
  ) %>%
```

```r
  kable_styling(latex_options = c("striped", "hold_position"),
                full_width = FALSE)
```

Table 2: Player Engagement Metrics by Game Genre

Genre	Avg Play Time (min)	Avg Score	Sessions	Unique Players
Puzzle	91.67	496.52	6371	3606
Sports	92.59	503.38	3995	2769
Strategy	92.88	500.84	3643	2599
Action	92.10	493.06	3199	2370
RPG	91.41	499.69	2792	2139

3.5 Visualisation

```r
ggplot(task2_summary,
       aes(x = reorder(genre, num_unique_players),
           y = num_unique_players, fill = genre)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = num_unique_players),
            hjust = -0.2, size = 4.2) +
  coord_flip() +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title    = "Number of Unique Players by Game Genre",
    subtitle = "Based on all recorded session data",
    x        = "Genre",
    y        = "Number of Unique Players"
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"))
```

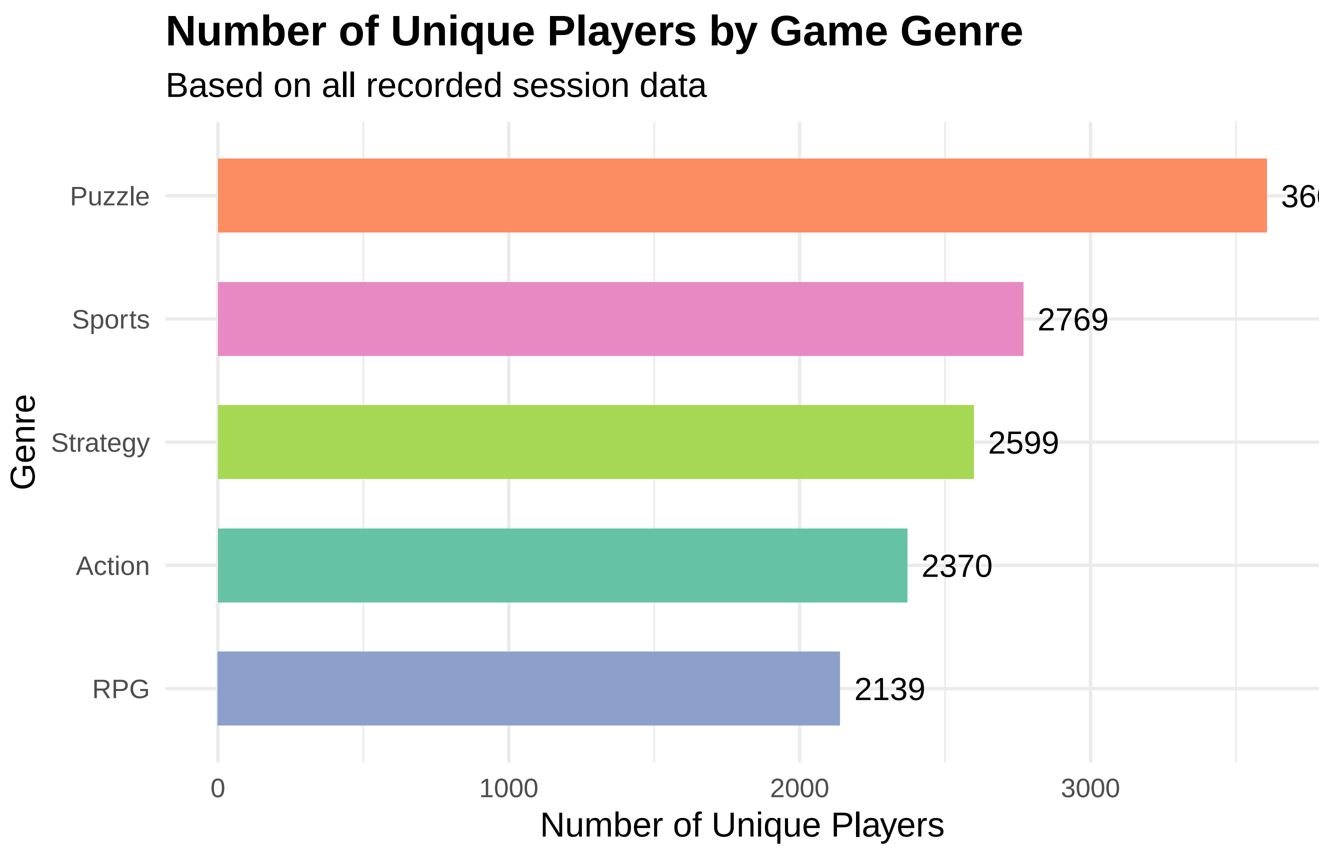

Figure 2: Number of Unique Players by Game Genre

3.6 Interpretation

Puzzle had the most unique players and sessions with 3,606 and 6,371 respectively compared to almost twice as many for RPG. RPG had the fewest unique players and sessions which was surprising as RPGs tend to be more popular. Sports and Strategy fell somewhere in the centre. Strategy did not have the most unique players but players who played Strategy had the highest number of sessions with the game compared to all other genres indicating that the players who played Strategy were more interested in the game. Sports earned the highest number of sessions with players scoring games in Sports. Given these results, it would be beneficial for the company to continue investing in Puzzle as it has the greatest number of unique players but also potentially investigate why players are so interested in Strategy as this could potentially be applied to the RPG genre to also increase its number of players.

4 Task 3 — Top Players and Their Behaviour

4.1 Task Explanation

For this task I had to find the top 10 players based upon the total amount of time they spent playing the game. For these top 10 players I looked at how many games per week they played on average alongside their average game score. I wondered if these two variables correlated to one another - players who played the most having the highest scores for the game.

4.2 Assumptions

I positioned players according to their total cumulative play time - the sum of all their sessions. Any player with NA values for play time I excluded those sessions from the total play time ranking. I took all sessions for the behavioural metrics for each of the top 10 players, with NAs dealt with via na.rm.

4.3 Data Wrangling

```r
# Step 1: Rank all players by total play time and keep the top 10
top10_player_ids <- sessions %>%
  filter(!is.na(play_time_minutes)) %>%       # exclude any NA play times
  group_by(player_id) %>%
  summarise(total_play_time = sum(play_time_minutes)) %>%
  slice_max(total_play_time, n = 10) %>%      # top 10 by cumulative play time
  pull(player_id)                             # extract just the IDs

# Step 2: Filter sessions to top 10 players and compute behavioural metrics
task3_summary <- sessions %>%
  filter(player_id %in% top10_player_ids) %>%
  group_by(player_id) %>%
  summarise(
    total_sessions    = n(),
    avg_play_time_min = round(mean(play_time_minutes, na.rm = TRUE), 2),
    avg_score         = round(mean(score, na.rm = TRUE), 2)
  ) %>%
  arrange(desc(avg_play_time_min))
```

4.4 Results Table

```r
task3_summary %>%
  kable(
    col.names = c("Player ID", "Total Sessions",
                  "Avg Play Time (min)", "Avg Score"),
    caption   = "Behaviour Summary of Top 10 Players by Total Play Time",
```

```r
    booktabs   = TRUE
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position"),
                full_width = FALSE)
```

Table 3: Behaviour Summary of Top 10 Players by Total Play Time

Player ID	Total Sessions	Avg Play Time (min)	Avg Score
3332	9	133.11	528.00
3391	9	130.11	535.11
3812	9	129.67	429.11
2809	10	119.40	418.50
4840	10	115.60	433.50
2447	13	115.54	617.69
2423	13	106.85	452.46
103	11	104.91	438.18
4175	13	99.85	504.31
2135	12	95.92	555.83

4.5 Visualisation

```r
sessions %>%
  filter(player_id %in% top10_player_ids) %>%
  mutate(player_id = factor(player_id)) %>%    # treat ID as a categorical variable
  ggplot(aes(x = player_id, y = score, fill = player_id)) +
    geom_boxplot(show.legend = FALSE,
                 outlier.colour = "red", outlier.shape = 16, outlier.size = 2) +
    scale_fill_brewer(palette = "Paired") +
    labs(
      title    = "Score Distribution for Top 10 Players",
      subtitle = "Players ranked by cumulative play time; red dots = outliers",
      x        = "Player ID",
      y        = "Score"
    ) +
    theme_minimal(base_size = 13) +
    theme(plot.title  = element_text(face = "bold"),
          axis.text.x = element_text(angle = 45, hjust = 1))
```

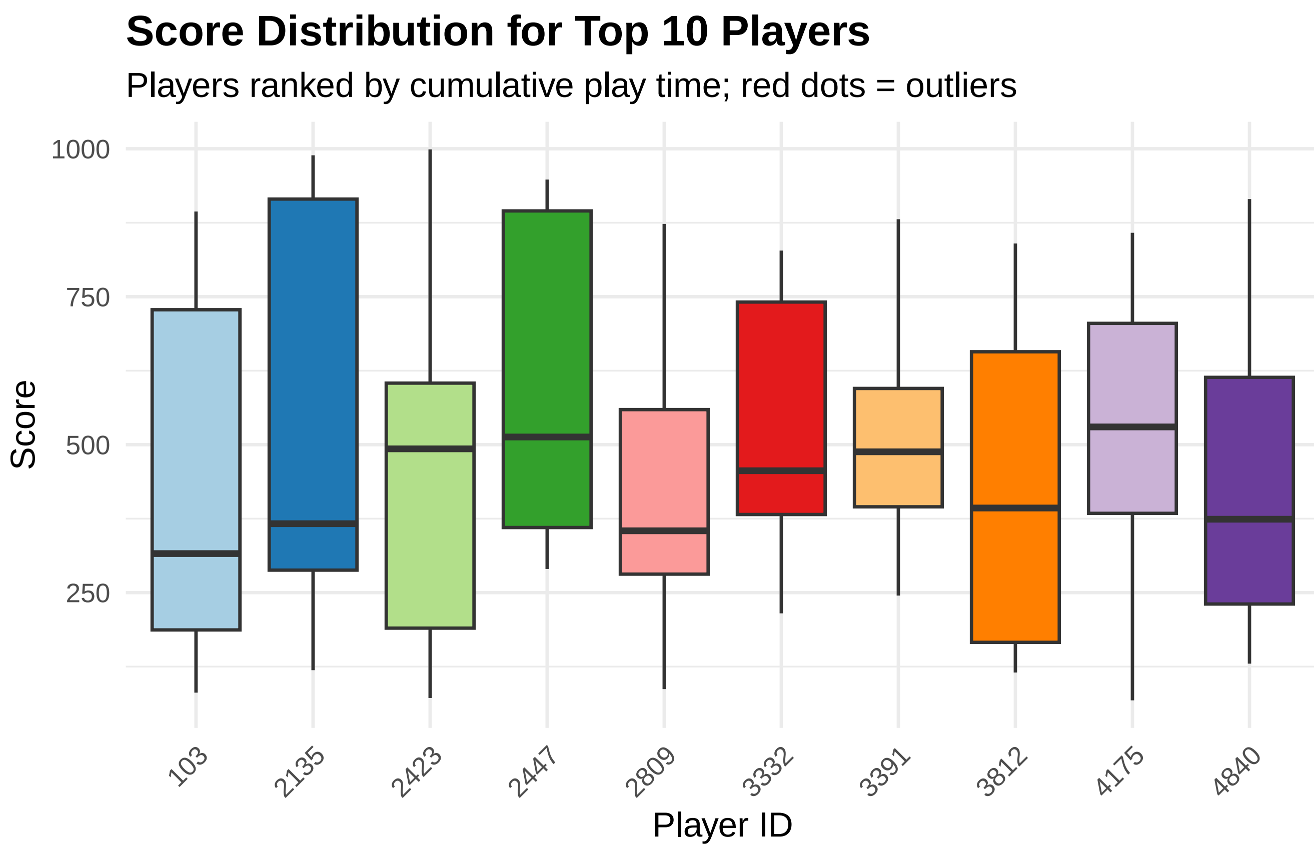

Figure 3: Score Distribution for the Top 10 Players (Boxplot)

4.6 Interpretation

These 10 players are very much above average - the very top one (ID 3332) sees 133 minutes per session - nearly 40 minutes above the platform average. All have session counts between 9 and 13 though so each returns regularly. What is interesting though is that play time and score do not necessarily match up. Player 2447 features an average score per session that is very high at 617 while player 3812 puts in some of the longest sessions but only averages 429. Such gap indicates that some of these top players are putting in efforts without necessarily being the best players - perhaps they explore the games in a more casual manner or try out various strategies instead of optimising for score.

5 Task 4 — Age Group Behaviour Comparison

5.1 Task Explanation

For this task I split the players into four age groups and compared how each behaved on the platform - calculating average play time, score and total sessions per group to reveal whether younger players are indeed more active or better performers - which is what one would probably assume going in

- **16–25**: Youngest players
- **26–35**: Young adults
- **36–45**: Middle-aged players
- **46+**: Older players

5.2 Assumptions

I used each player's current recorded age to form the groups. Any players with a missing age value were left out from the brackets as there's no way to assign them to this bracket. All gaming sessions contribute to the statistics of each age group.

5.3 Data Wrangling

```r
# Step 1: Assign each player to an age bracket
players_age <- players %>%
  filter(!is.na(age)) %>%          # exclude players with missing age
  mutate(
    age_group = case_when(
      age >= 16 & age <= 25 ~ "16-25",
      age >= 26 & age <= 35 ~ "26-35",
      age >= 36 & age <= 45 ~ "36-45",
      age >= 46              ~ "46+",
      TRUE                   ~ NA_character_   # catch unexpected values
    )
  ) %>%
  filter(!is.na(age_group))        # remove any unclassified players

# Step 2: Join age groups with sessions to attach age info to each session
sessions_age <- players_age %>%
  select(player_id, age_group) %>%
  inner_join(sessions, by = "player_id")

# Step 3: Summarise gaming behaviour per age group
task4_summary <- sessions_age %>%
  group_by(age_group) %>%
  summarise(
    avg_play_time_min = round(mean(play_time_minutes, na.rm = TRUE), 2),
```

```
    avg_score          = round(mean(score, na.rm = TRUE), 2),
    num_sessions       = n()
  ) %>%
  # Set factor levels to display age groups in ascending order
  mutate(age_group = factor(age_group,
         levels = c("16-25", "26-35", "36-45", "46+"))) %>%
  arrange(age_group)
```

5.4 Results Table

```
task4_summary %>%
  kable(
    col.names = c("Age Group", "Avg Play Time (min)",
                  "Avg Score", "Number of Sessions"),
    caption   = "Gaming Behaviour Metrics by Age Group",
    booktabs  = TRUE
  ) %>%
  kable_styling(latex_options = c("striped", "hold_position"),
                full_width = FALSE)
```

Table 4: Gaming Behaviour Metrics by Age Group

Age Group	Avg Play Time (min)	Avg Score	Number of Sessions
16-25	91.80	496.69	4413
26-35	91.57	503.53	4538
36-45	91.58	498.02	4756
46+	93.11	496.71	6293

5.5 Visualisation

```
ggplot(task4_summary,
       aes(x = age_group, y = avg_play_time_min, fill = age_group)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = avg_play_time_min),
            vjust = -0.4, size = 4.5, fontface = "bold") +
  scale_fill_manual(values = c("16-25" = "#66C2A5",
                               "26-35" = "#FC8D62",
                               "36-45" = "#8DA0CB",
                               "46+"   = "#E78AC3")) +
  labs(
    title    = "Average Play Time by Age Group",
    subtitle = "Comparison of session duration across four age brackets",
```

```r
    x         = "Age Group",
    y         = "Average Play Time (minutes)"
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"))
```

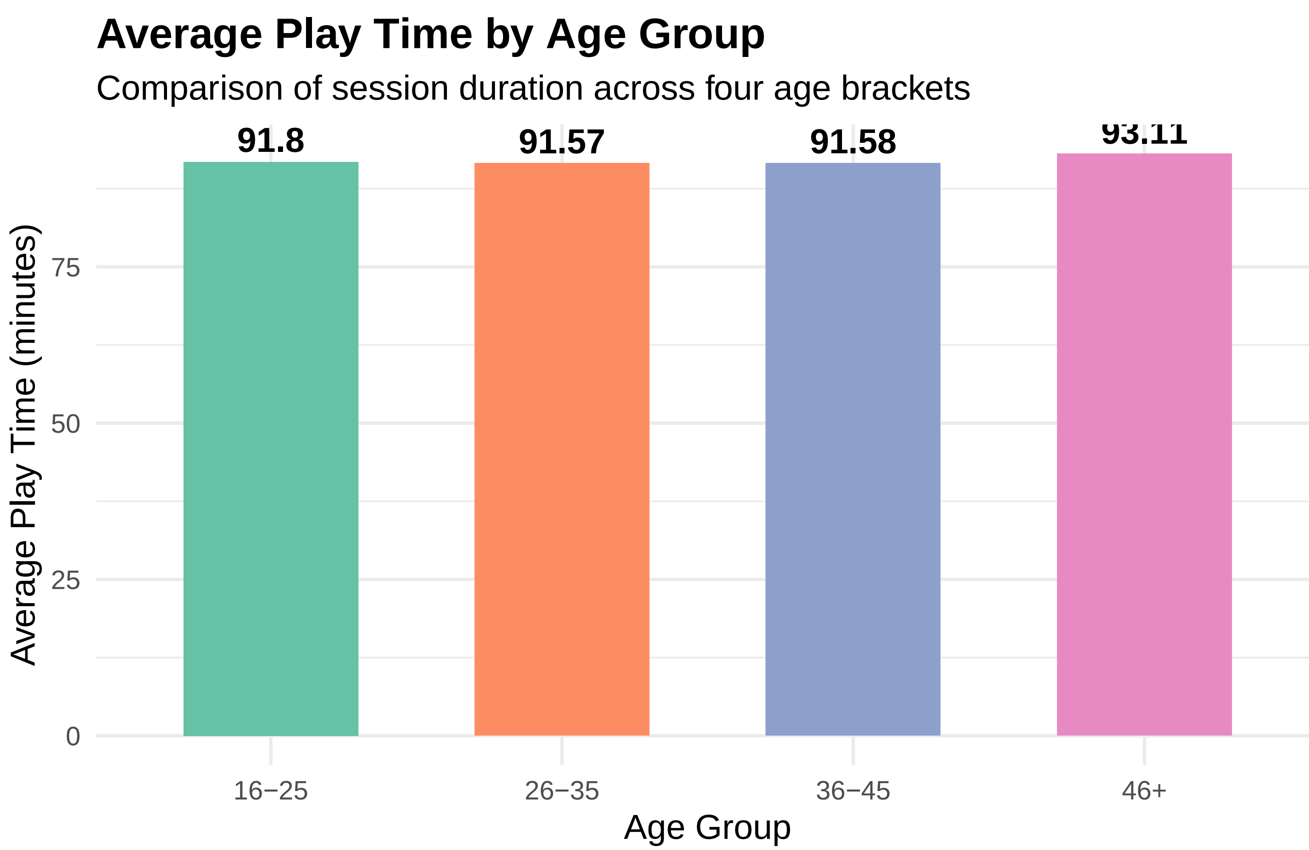

Figure 4: Average Play Time by Age Group

5.6 Interpretation

The results did create a bit of a flip in the expectations. The 46+ demographic included both the longest average sessions (93.11 min) and the most total sessions - making this the most active of the groups on the platform. The 16-25 demographic featured the fewest sessions within all four groups. The scores were pretty even though (496-503) so no age group is dramatically outperforming the others in terms of skill. The 26-35 group edged slightly ahead with the average score (503.53) but that's not a huge difference. Overall then, Play Pulse Studios probably shouldn't focus too much of their marketing on younger audiences - the older players are very engaged and loyal customers - and hence worth targeting even more directly.

6 Summary

Looking across all four tasks, a few things stand out. The level of experience among players had almost no effect on how they played the game - veterans and new players exhibit identical behaviour in terms of session length and score. Puzzle proved to be the most popular genre with the largest base of players whilst Strategy deserves attention due to the length of players' sessions. The top ten players reveals that long playing times don't necessarily translate into high performance - scores show quite a spread amongst the heaviest users of the games. Most surprisingly though, the 46+ age group is the most active upon the platform - challenging the perception of gaming as mainly a young person's hobby. The platform as a result has very broad appeal across the variety of player types - a solid position for the studio to be in.